

Agile Development I: The Manifesto and Scrum

Why agile is an economic argument rather than a fashion, and how a four-student team runs a two-week sprint — roles, artifacts, events and a Definition of Done, all instantiated for SANTE-BASSAM.

**DATE**

Tuesday, 22 September 2026

MEETING TIME

1:00 - 2:20 PM

ROOM FORMAT

Lecture + sprint-planning exercise

05

2 concepts worked on SANTE-BASSAM

11 Scrum elements instantiated

16 backlog stories

5 figures

0

The session in one picture

Agile is not "faster" or "more modern". It is a rational response to **one economic condition: cheap feedback**. When you can put working software in front of a real user every two weeks, the cheapest way to find the right requirement is to build a small wrong one and correct it. When you cannot, that argument collapses.

FIG. 0 · ONE QUESTION DECIDES THE METHOD: HOW MUCH DOES IT COST TO SHOW THE USER SOMETHING THAT WORKS?

FEEDBACK EXPENSIVE

FEEDBACK CHEAP

A satellite · an implanted device

You cannot show a half-built satellite to the customer.
Every wrong guess is discovered after launch.

→ Spend heavily on getting the specification right first.

PLAN-DRIVEN

Session 3: waterfall, V, spiral

SANTE-BASSAM's front desk

The post is 20 min from campus; the nurse reviews fortnightly.
A wrong screen costs one sprint to fix, not one project.

→ Build small, show it Friday, correct it Monday.

AGILE

Today: Scrum · Friday: XP, stories, Kanban

Exam trap: "agile is faster" ✗ — "agile is rational when feedback is cheap" ✓

Session 4 ended with a line drawn through SANTE-BASSAM: what the law and the architecture fix above it, what the users must teach you below it. Today supplies the mechanics for *below the line*.

1 Learning outcomes

L0 5.1 State the four Agile Manifesto values and the economic condition — cheap feedback — under which each trade is rational.

L0 5.2 Describe Scrum's 3 roles, 3 artifacts and 5 events, and say what each one inspects.

L0 5.3 Instantiate all eleven elements for the SANTE-BASSAM team, including a written Definition of Done.

L0 5.4 Order a product backlog by risk and value, and write a sprint goal a stakeholder can verify on a named date.



How this session is built. The Manifesto and Scrum are worked in full on SANTE-BASSAM today. XP, user stories and Kanban follow in Session 6.

2 Where the project stands



THE SYSTEM

SANTE-BASSAM

Registry + appointments, ~6,000 people, one laptop, one tablet, weekly power cuts.



FROM SESSION 3

Sprint 0 = the spike

Offline capture and sync proven in weeks 1-4. R1 and R4 retired.



CARRIED FORWARD

UR-1..3 · SR-1.1..3.1

Today they reappear as stories S-01..S-16 — the same needs, in a form a sprint can consume.



THE CLOCK

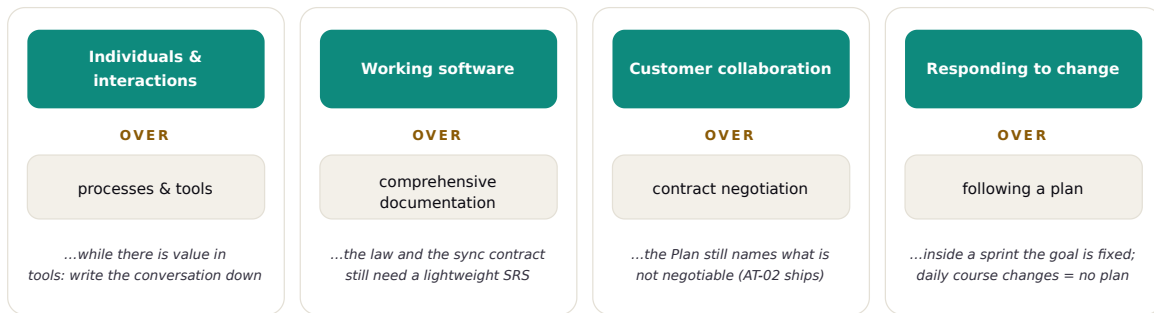
Weeks 5-14 · five sprints

Ten weeks to turn a working sync path into a system the post runs alone.

3.1 The Manifesto as economics

Snowbird, Utah, February 2001: seventeen practitioners, four value statements, each of the form *we value X over Y — while there is value in Y*. Drop the second clause and you get "no documentation, no planning", which is how agile earned its bad reputation.

FIG. 1 · FOUR TRADES — AND THE CLAUSE EVERYONE FORGETS TO QUOTE



APPLIED TO SANTE-BASSAM

The four trades, priced for your team — is feedback cheap here? Yes, for the workflow.

INDIVIDUALS OVER TOOLS · WHAT IT BUYS

The clerk says the search screen is confusing → two students go to the desk on Friday with the tablet. Not a ticket.

What you still owe: a one-paragraph note in the repo the same day. Interactions that leave no trace are not engineering.

WORKING SOFTWARE OVER DOCUMENTS

Week 6: the clerk types into a registration screen, not a 40-page SRS. Her reaction *is* the requirement.

What you still owe: the lightweight SRS (Session 10) — identifier format, consent, sync contract cannot be discovered by typing into a screen.

COLLABORATION OVER CONTRACT

Week 8: the nurse asks for a follow-up list → into the backlog; something else moves down.

What you still owe: the Plan names the non-negotiable — AT-02 (nothing lost in a power cut) ships in 1.0 whatever else is cut.

CHANGE OVER PLAN

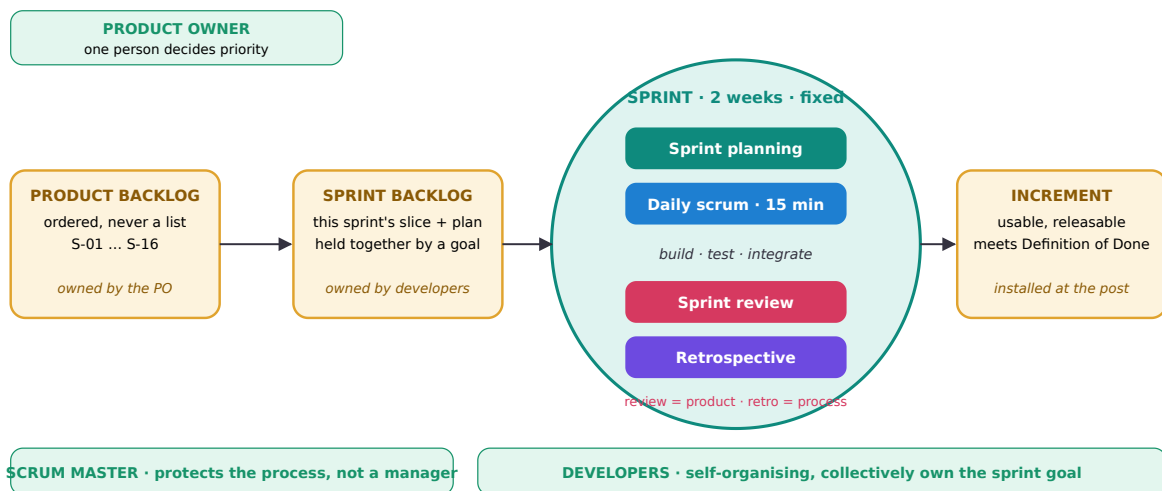
Week 5's paper prototype shows search-by-mother is essential → it enters Sprint 3; card printing leaves.

What you still owe: a fixed goal *inside* the sprint. Changing course every day is not responsiveness.

3.2 Scrum: a management framework, not an engineering one

One rule: Scrum schedules feedback. The review inspects the product; the retrospective inspects the process. None of its five events says how to write code — which is why teams adopt Scrum, keep bad engineering habits, and conclude agile doesn't work.

FIG. 2 · THE SCRUM LOOP — 3 ROLES, 3 ARTIFACTS, 5 EVENTS



The most under-appreciated element is the **Definition of Done**: the team's written standard for "complete". Without it, velocity is fiction, because each item is done to a different standard.

All eleven elements, instantiated for four students and one external customer

ROLES

PRODUCT OWNER

One student, proxy for the nurse

The nurse can't attend planning; someone orders the backlog daily on her behalf and answers to her fortnightly. She has the final say at each review.

SCRUM MASTER

A second student — rotates each sprint

Rotation stops one person becoming the manager; everyone learns to run a retrospective.

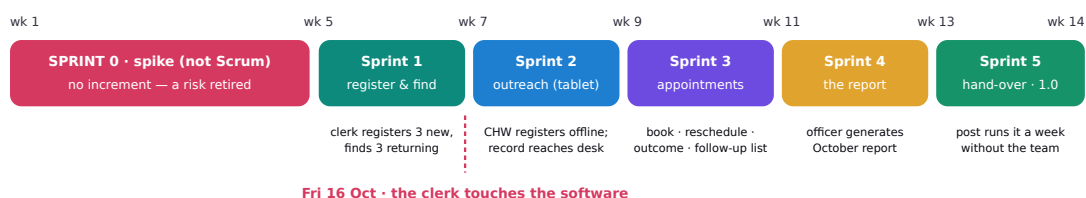
DEVELOPERS

All four, PO and SM included

A four-person team cannot afford two non-coding roles.

THE CALENDAR — FIVE SPRINTS IN FOURTEEN WEEKS

FIG. 3 · EACH SPRINT HAS A GOAL THE CLERK, CHW, NURSE OR OFFICER CAN VERIFY



THE PRODUCT BACKLOG AFTER SPRINT 0 — ORDERED BY THE SESSION 4 RISK REGISTER, NOT BY EASE OF BUILDING

#	AS A ... I WANT ... SO THAT ...	TRACES TO	PTS	SPRINT
S-01	team · journaled local store on both devices · no record lost when power fails	SR-2.1	5	done (0)
S-02	team · locally generated patient ids · two devices register offline without collision	SR-2.1	3	done (0)
S-03	clerk · paper register migrated · returning patients already exist	UR-1	5	done (0)

#	AS A ... I WANT ... SO THAT ...	TRACES TO	PTS	SPRINT
S-04	clerk · register a new patient with the paper-card fields · stop writing it twice	SR-1.1	5	1
S-05	clerk · find a patient by mother's name + village · serve mothers with no card	SR-1.1, 1.2	8	1
S-06	clerk · search result under 2 s · the queue does not grow	SR-1.3	3	1
S-07	CHW · register while offline · enrol families during outreach	SR-2.2	5	2
S-08	nurse · offline records reach the desk in creation order · the register stays consistent	SR-2.2	5	2
S-09	nurse · records failing sync 3× flagged for review · nothing silently vanishes	SR-2.3	3	2
S-10	clerk · book an appointment on a date · the nurse can plan her day	new (UR-4)	5	3
S-11	nurse · record each visit attended / missed / rescheduled · the report is true	SR-3.1	3	3
S-12	nurse · list of patients due for follow-up this week · retire my notebook	new (UR-5)	5	3
S-13	officer · monthly report by village and outcome · stop counting the register	SR-3.1	5	4
S-14	officer · report exported as a spreadsheet · check two villages by hand	SR-3.1	3	4
S-15	clerk · warning when a registration looks like an existing patient · no duplicates	R7	5	4
S-16	clerk · print a patient card · the patient leaves with something	nice-to-have	3	—

SPRINT 1 GOAL & BACKLOG

"On Friday 16 October the clerk registers three new patients and finds three returning ones on the desk app, unaided."

S-04 + S-05 + S-06 = 16 points — what the team actually completed in Sprint 0. Tasks under S-05: search screen · ranking function (UT-02 first) · prefix index · ST-01/ST-02 scripted · paper-prototype session with the clerk on day 3, before the screen is built.

DEFINITION OF DONE — IN THE REPO, AGREED BY ALL FOUR

- Merged to `main` via a PR reviewed by someone who did not write it
- Unit tests for every new function; whole suite green in CI on the merge commit
- Acceptance criteria demonstrated *on the target device*
- New fields in the data dictionary; new interfaces in the sync contract
- Traceability matrix updated: story → requirement → tests
- Demonstrable to the nurse without the developer explaining what to click

THE FIVE EVENTS, DATED

PLANNING · MON WK 5 · 90 MIN

Goal, then tasks

PO proposes the goal; developers pull S-04..06 and break them into tasks. Nobody leaves until the tasks fit two weeks.

DAILY · 12:45 · 15 MIN · STANDING

Re-plan the day

Finished / today / blocked. The SM writes the blocker down. Nothing else is discussed there.

REVIEW · FRI WK 6 · AT THE POST

The clerk uses it

The team watches and writes; it does not explain. The nurse says what to change.

RETRO · SAME FRIDAY · CAMPUS

Keep one, change one

The change enters Sprint 2 as a task with an owner.

SPRINT · WEEKS 5–6

Fixed

If S-06 can't finish, it returns to the backlog. The sprint does not extend.



What Scrum gave the team: a date — Friday 16 October — on which the clerk touches the software. Session 3's waterfall team first met the clerk in week 12. That is the whole difference; Scrum is the machinery that makes the early date happen every fortnight.

4

In-class exercise · plan Sprint 2

Groups of four, 15 minutes. Sprint 1 is planned in Fig. 3 and the backlog. Do the same for Sprint 2 — the tablet sprint.



Goal sentence

— one sentence, with a date and a person: "On Friday ... the CHW can ...". It must be verifiable at the post without the team explaining anything.



Select stories

— from the backlog, to a total the team can finish (Sprint 0 says 16 points). Say what you left out and why.



Break down the largest story

— five or six tasks, the test that is written first, and the day the CHW touches it.



One DoD change

— which item of the Definition of Done does the tablet sprint make insufficient? Rewrite it.



Expect this. Most groups pick S-07, S-08 and S-09 (13 points) and forget that "demonstrable on the target device" now means a tablet in a village with no network — the DoD item that changes is the demonstration one.

5

Homework · due before Session 6 (Friday 25 September)



Product backlog

`backlog.md` — ≥ 15 stories in S-xx format: role, capability, benefit, points, trace to a UR/SR or *new*. Ordered. At least three from stakeholders other than the clerk.



Definition of Done

`dod.md` — start from today's; change at least one item and say why.



Sprint 1 plan

— goal sentence with a date; stories totalling what four students can finish in two weeks; tasks for the largest story.



Reading

— Sommerville Ch. 3, §3.1–3.2; *The Scrum Guide* (14 pages).



Assessment link. The backlog is the evidence base for Session 7's elicitation: every story is a hypothesis about what someone needs.

TWO IDEAS · ONE RULE EACH · WHAT EACH GAVE SANTE-BASSAM

MANIFESTO

X over Y — while Y still has value

Rational when feedback is cheap.
The post is 20 minutes away and the nurse reviews fortnightly — so yes, for the workflow. Not for the law.

SCRUM

schedules feedback, not engineering

3 roles, 3 artifacts, 5 events, a fixed sprint and a written Definition of Done. Result: Friday 16 October, the clerk touches the software.

FRIDAY

XP · stories · Kanban

The engineering half Scrum leaves out, the unit of work a sprint consumes, and the discipline of flow.

If you remember one sentence: **agile is what a team does when it can afford to be wrong cheaply.**

Key terms

cost of feedback

Agile Manifesto

Scrum

Product Owner

sprint goal

Definition of Done

increment

velocity

Readings and references

1. Sommerville, I. *Software Engineering*, 10th ed. — Chapter 3.
2. Beck, K. et al. *Manifesto for Agile Software Development*, 2001 — including the twelve principles.

3. Schwaber, K. & Sutherland, J. *The Scrum Guide*, 2020.

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Next · Session 06 — Agile Development II: XP, User Stories and Kanban